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 Studierichting: Informatica
 Oefeningenreeks 5D nr. 4.8

$$y = x^3 - 3x + 2, y = 0$$

$$\Rightarrow x^3 - 3x + 2 = 0 \Rightarrow (x - 1)(x^2 + x - 2) = 0 \Rightarrow (x - 1)^2(x + 2) = 0$$

$$\begin{aligned} S &= \int_{-2}^1 (x^3 - 3x + 2) dx \\ &= \left[\frac{x^4}{4} - \frac{3x^2}{2} + 2x \right]_{-2}^1 \\ &= \frac{1}{4} - \frac{6}{4} + 2 - 4 + 6 + 4 = \frac{27}{4} \end{aligned}$$

Tekening:

x	$-\infty$	-2			-1	1			$+\infty$
$f(x)$		$-$	0	$+$	4	$+$	0	$+$	
$f'(x)$		$+$	$+$	$+$	0	$-$	0	$+$	

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References